

# A Provenance-Preserving Quotient of Literal Affine Maps:

Exact Collision Splitting and the Identical-Map Concentration Ledger

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## Abstract

TPC-100 linearizes every remaining invertible opened atom as

$$\omega_p = A_p(m_p j_p + h_0) \quad \text{in } \mathbb{F}_{q_X}^\times,$$

but its maximally literal fine cells are singletons. Thus all distinct-atom collisions remain in the cross-cell ledger, and different cells may induce the same affine map. We construct the canonical quotient required before a genuine cross-map estimate is attempted.

Attach to an atom  $p$  the map

$$\phi_p(x) = s_p x + t_p, \quad s_p = A_p m_p, \quad t_p = A_p h_0.$$

Atoms are equivalent when their pairs  $(s_p, t_p)$  agree. A lossless quotient record consists of the full recoverable provenance bag together with the input profile

$$\nu_\phi(j) = \sum_{\substack{p: \phi_p = \phi \\ j_p = j}} w_p.$$

The profile, not merely the total map weight, is necessary and sufficient for the positive multiplier census:

$$a_H^{\text{inv}}(\omega) = \sum_{\phi} \sum_j \nu_\phi(j) \mathbf{1}_{\{\phi(j)=\omega\}}.$$

We give a two-point counterexample proving that map totals alone lose the census.

Because  $h_0, A_p, m_p$  are nonzero modulo the strong no-wrap prime, two maps agree exactly when  $A_p = A_{p'}$  and  $m_p = m_{p'}$  modulo  $q_X$ ; no-wrap upgrades the latter to equality of the physical row integers. Within one map, output collision is exactly equality of the input coordinates. Consequently the TPC-100 centered cross ledger splits identically as

$$\mathfrak{X}_H^\circ = \mathfrak{K}_H^\circ + \mathfrak{Y}_H^\circ,$$

where  $\mathfrak{K}_H^\circ$  contains identical maps and  $\mathfrak{Y}_H^\circ$  contains genuinely distinct maps.

If  $M_{H,\phi}$  is the mass of one map class,  $U_X = \max_{H,\phi} M_{H,\phi}$ , and  $M_H$  is the total width mass, then

$$|\mathfrak{K}_H^\circ| \leq U_X M_H,$$

and therefore

$$\sum_H g_{q_X}(H) \sqrt{|\mathfrak{K}_H^\circ|} \leq \sqrt{U_X |\mathcal{H}_X| (q_X - 1) \mathfrak{P}_X}.$$

This is an exact L1 reduction. It exposes a new identical-map concentration ledger  $U_X$  and leaves the width-weighted distinct-map term as the honest TPC-106 geometric door. Neither ledger is bounded here at the growing physical scale; no L2 arithmetic saving or parity breakthrough is claimed.

**Keywords:** affine map; provenance quotient; collision energy; identical maps; incidence; fixed shift; positive resonance.

**Literal-frame convention.** Every quotient starts from the actual  $q_X \nmid u$  opened atoms, keeps one prescribed  $h_0 \neq 0$ , their full source and child records, actual weights and support, both polarizations, and the single inherited physical normalization. Nothing is specialized to  $h_0 = 2$ .

## 1 Why a quotient is needed

Fix an exact resonance width  $H$ . TPC-100 partitions the actual invertible opened atoms into maximally literal fine cells. Because each cell retains  $m, u, \sigma$ , the source identity

$$mj + h_0 = \sigma u$$

recovers  $j$ ; every fine cell is therefore a singleton. The resulting multiplier census is

$$a_H^{\text{inv}}(\omega) = \sum_{\substack{p: \mathbf{H}_p = H \\ q_X \nmid u_p}} w_p \mathbf{1}_{\{\omega_p = \omega\}}, \quad (1)$$

where  $w_p \geq 0$  is the exact physical atom weight [1].

The centered energy has an atomic diagonal part and a cross part

$$\|(a_H^{\text{inv}})^\circ\|_2^2 = \mathfrak{D}_H^\circ + \mathfrak{X}_H^\circ.$$

TPC-103 closes the maximum atom cap, so the diagonal part follows from the unresolved principal mass [2, 4]. The cross part remains. Before applying an incidence theorem to it, TPC-102 and TPC-MVP1 require identical affine maps to be aggregated without losing source recovery or exact weight [3, 7].

The subtle point is that an affine map is an operator on an input, not merely a multiplier value. Two atoms may carry the same map but occur at different  $j$ 's. Their total class mass does not determine where that mass is sent.

## 2 Literal affine maps

Write  $q = q_X$ . For all sufficiently large  $X$ , the strong no-wrap choice gives

$$q > |h_0|, \quad 0 < m_p < q, \quad q \nmid u_p.$$

TPC-100 defines

$$A_p = \varepsilon_p \ell_p v_p B_p (c_p u_p)^{-1} \in \mathbb{F}_q^\times$$

and proves

$$\omega_p = A_p(m_p j_p + h_0). \quad (2)$$

**Definition 2.1** (Literal affine-map record). For an active atom  $p$ , define

$$s_p = A_p m_p, \quad t_p = A_p h_0, \quad \phi_p(x) = s_p x + t_p.$$

Thus  $s_p, t_p \in \mathbb{F}_q^\times$  and  $\omega_p = \phi_p(j_p)$ .

**Lemma 2.2** (Exact identical-map classification). *For two active atoms  $p, p'$ ,*

$$\phi_p = \phi_{p'} \iff A_p = A_{p'} \text{ and } m_p = m_{p'} \text{ in } \mathbb{F}_q.$$

*Under the strong no-wrap bounds, equality of  $m_p, m_{p'}$  in  $\mathbb{F}_q$  is equality as physical integers.*

*Proof.* Equality of the constant terms gives

$$h_0(A_p - A_{p'}) = 0.$$

Since  $q \nmid h_0$ , this forces  $A_p = A_{p'}$ . Equality of the slopes and invertibility of  $A_p$  then force  $m_p = m_{p'}$ . The converse is immediate. Finally both physical row integers lie strictly between 0 and  $q$  [6].  $\square$

**Corollary 2.3** (Collision inside one map). *If  $\phi_p = \phi_{p'}$ , then*

$$\omega_p = \omega_{p'} \iff j_p = j_{p'} \pmod{q}.$$

*Since the actual orbit window has diameter below  $q$ , this is equivalent to equality of the physical integers  $j_p = j_{p'}$ .*

*Proof.* The common slope  $s_p$  is nonzero, so  $\phi_p(j_p) = \phi_{p'}(j_{p'})$  exactly when  $j_p = j_{p'}$  modulo  $q$ . Apply no-wrap.  $\square$

*Remark 2.4* (What identical maps may still hide). Equality of  $(A, m, j)$  does not identify the full atom. Different content or projector children, complementary factorizations, masks, polarizations, or outer source labels may still contribute separate weights. Those records may not be discarded.

### 3 The canonical provenance quotient

Let  $\mathcal{P}_H$  be the finite set of active invertible atoms at width  $H$ . Define

$$p \sim p' \iff (s_p, t_p) = (s_{p'}, t_{p'}).$$

Let  $\Phi_H$  be the set of equivalence classes, identified with their common affine maps.

**Definition 3.1** (Provenance bag and input profile). For  $\phi \in \Phi_H$ , let

$$\mathcal{B}_\phi = \{(p, w_p, \mathbf{r}_p) : \phi_p = \phi\},$$

where  $\mathbf{r}_p$  is the complete TPC-100 augmented source and child record. Define

$$\nu_\phi(j) = \sum_{\substack{p \in \mathcal{P}_H \\ \phi_p = \phi, j_p = j}} w_p, \tag{3}$$

$$M_{H,\phi} = \sum_j \nu_\phi(j), \quad M_H = \sum_{\phi \in \Phi_H} M_{H,\phi}. \tag{4}$$

The canonical quotient record is

$$\mathcal{Q}_H = \{(\phi, \nu_\phi, \mathcal{B}_\phi) : \phi \in \Phi_H\}.$$

**Theorem 3.2** (Lossless quotient and exact census). *The classes  $\mathcal{B}_\phi$  form a disjoint partition of the actual atoms, preserve every physical weight exactly once, and give*

$$a_H^{\text{inv}}(\omega) = \sum_{\phi \in \Phi_H} \sum_{j \in \mathbb{F}_q} \nu_\phi(j) \mathbf{1}_{\{\phi(j)=\omega\}}. \tag{5}$$

*The analytic data  $(\phi, \nu_\phi)$  recover the complete positive multiplier census, while the attached bags recover every original source atom and coefficient.*

*Proof.* Every atom has one and only one pair  $(s_p, t_p)$ , so the bags form a partition. Reindex (1) first by that pair and then by  $j_p$ . This gives (5) and

$$\sum_{\phi, j} \nu_\phi(j) = \sum_p w_p = M_H.$$

No weight is created or deleted. Source recovery follows because the full augmented record  $\mathbf{r}_p$  remains in the corresponding bag.  $\square$

**Proposition 3.3** (Map totals alone are not lossless). *The collection  $\{(\phi, M_{H,\phi})\}$  does not determine  $a_H^{\text{inv}}$ , even for one invertible affine map and one unit atom.*

*Proof.* Take  $q \geq 5$ ,  $h_0 = 1$ , and  $\phi(x) = x + 1$ . One unit atom at  $j = 0$  gives the census  $\delta_1$ , while one unit atom at  $j = 1$  gives  $\delta_2$ . Both quotient records have the same map and total mass one. Their input profiles, and hence their multiplier censuses, differ.  $\square$

*Remark 3.4* (Minimal analytic payload). The proposition shows why “aggregate identical maps with exact total weight” is incomplete unless the input distribution is also retained. The profile  $\nu_\phi$  is sufficient for every positive incidence calculation. The provenance bag is additionally required for literal source recovery and later signed reconstruction.

## 4 Exact splitting of the cross ledger

Since the TPC-100 fine cells are singleton, its centered cross ledger can be written directly as

$$\mathfrak{X}_H^\circ = \sum_{\substack{p, p' \in \mathcal{P}_H \\ p \neq p'}} w_p w_{p'} \left( \mathbf{1}_{\{\omega_p = \omega_{p'}\}} - \frac{1}{q-1} \right). \quad (6)$$

The sum is ordered, matching the  $C \neq D$  convention in TPC-100.

**Definition 4.1** (Identical- and distinct-map ledgers). Let

$$\mathfrak{K}_H^\circ = \sum_{\substack{p \neq p' \\ \phi_p = \phi_{p'}}} w_p w_{p'} \left( \mathbf{1}_{\{\omega_p = \omega_{p'}\}} - \frac{1}{q-1} \right), \quad (7)$$

$$\mathfrak{Y}_H^\circ = \sum_{\substack{p, p' \\ \phi_p \neq \phi_{p'}}} w_p w_{p'} \left( \mathbf{1}_{\{\omega_p = \omega_{p'}\}} - \frac{1}{q-1} \right). \quad (8)$$

**Theorem 4.2** (Canonical collision split). *One has the exact identity*

$$\boxed{\mathfrak{X}_H^\circ = \mathfrak{K}_H^\circ + \mathfrak{Y}_H^\circ.} \quad (9)$$

Moreover, put

$$S_{\phi,j} = \sum_{\substack{p: \phi_p = \phi \\ j_p = j}} w_p^2, \quad S_\phi = \sum_j S_{\phi,j}.$$

Then the identical-map term has the profile formula

$$\boxed{\mathfrak{K}_H^\circ = \sum_{\phi,j} \left( \nu_\phi(j)^2 - S_{\phi,j} \right) - \frac{1}{q-1} \sum_\phi \left( M_{H,\phi}^2 - S_\phi \right).} \quad (10)$$

The distinct-map term is

$$\boxed{\mathfrak{Y}_H^\circ = \sum_{\substack{\phi, \psi \in \Phi_H \\ \phi \neq \psi}} \sum_{j,k} \nu_\phi(j) \nu_\psi(k) \left( \mathbf{1}_{\{\phi(j) = \psi(k)\}} - \frac{1}{q-1} \right).} \quad (11)$$

*Proof.* The alternatives  $\phi_p = \phi_{p'}$  and  $\phi_p \neq \phi_{p'}$  partition every ordered pair  $p \neq p'$ , giving (9). Within one map, [corollary 2.3](#) replaces output equality by  $j_p = j_{p'}$ . At fixed  $(\phi, j)$ , the mass of ordered distinct pairs is

$$\nu_\phi(j)^2 - S_{\phi,j}.$$

Across all inputs of the same map, the total ordered distinct-pair mass is

$$M_{H,\phi}^2 - S_\phi.$$

This proves (10). Reindexing pairs from two different classes by their two input coordinates gives (11).  $\square$

*Remark 4.3* (Centering may couple the two terms). Neither  $\mathfrak{K}_H^\circ$  nor  $\mathfrak{Y}_H^\circ$  is necessarily nonnegative. Their exact sum is the native cross excess. Bounding their absolute values separately is a sufficient route and may lose cancellation; it is not asserted to be necessary.

## 5 The identical-map concentration ledger

**Definition 5.1** (Maximum map-class mass). Set

$$U_H = \max_{\phi \in \Phi_H} M_{H,\phi}, \quad U_X = \max_H U_H,$$

with zero used for an empty carrier.

**Theorem 5.2** (Identical-map concentration bound). *For every width  $H$ ,*

$$\boxed{|\mathfrak{K}_H^\circ| \leq \sum_{\phi \in \Phi_H} (M_{H,\phi}^2 - S_\phi) \leq U_H M_H.} \quad (12)$$

*Proof.* For one ordered pair, the centered collision kernel is either

$$1 - \frac{1}{q-1} \quad \text{or} \quad -\frac{1}{q-1},$$

so its modulus is at most one. Summing the weights of all ordered distinct pairs inside identical-map classes gives the first bound. For the second,

$$\sum_{\phi} (M_{H,\phi}^2 - S_\phi) \leq \sum_{\phi} M_{H,\phi}^2 \leq U_H \sum_{\phi} M_{H,\phi} = U_H M_H.$$

□

Let

$$\mathcal{H}_X = \{H : g_{q_X}(H) > 0, M_H > 0\}, \quad \mathfrak{P}_X = \sum_H \frac{2H}{q_X - 1} M_H.$$

**Corollary 5.3** (Width-weighted transfer). *The identical-map ledger obeys*

$$\boxed{\sum_H g_{q_X}(H) \sqrt{|\mathfrak{K}_H^\circ|} \leq \sqrt{U_X |\mathcal{H}_X| (q_X - 1) \mathfrak{P}_X}.} \quad (13)$$

Hence, if

$$q_X \asymp Q_X, \quad U_X \leq X^{o(1)}, \quad \mathfrak{P}_X \leq X^{o(1)} Q_X^2,$$

then the left side of (13) is  $X^{o(1)} Q_X^2$ .

*Proof.* By [theorem 5.2](#),

$$\sum_H g_q(H) \sqrt{|\mathfrak{K}_H^\circ|} \leq \sqrt{U_X} \sum_{H \in \mathcal{H}_X} g_q(H) \sqrt{M_H}.$$

Cauchy–Schwarz and  $g_q(H)^2 \leq 2H$  give

$$\leq \sqrt{U_X |\mathcal{H}_X| \sum_H 2H M_H} = \sqrt{U_X |\mathcal{H}_X| (q-1) \mathfrak{P}_X}.$$

Use  $|\mathcal{H}_X| \leq (q-1)/2$ .

□

**Corollary 5.4** (Exact sufficient distinct-map reduction). *Suppose the hypotheses in [corollary 5.3](#) hold and*

$$\sum_H g_{q_X}(H) \sqrt{|\mathfrak{Y}_H^\circ|} \leq X^{o(1)} Q_X^2. \quad (14)$$

*Then the complete width-weighted cross gate satisfies*

$$\sum_H g_{q_X}(H) \sqrt{|\mathfrak{X}_H^\circ|} \leq X^{o(1)} Q_X^2.$$

*Proof.* Use  $\mathfrak{X}_H^\circ = \mathfrak{K}_H^\circ + \mathfrak{Y}_H^\circ$  and

$$\sqrt{|x+y|} \leq \sqrt{|x|} + \sqrt{|y|}.$$

Then apply [corollary 5.3](#) and (14). □

## 6 Two exact nonimplications

**Proposition 6.1** (Atom cap does not imply map-class cap). *For every  $R \geq 1$ , there is an abstract positive opened census with*

$$W_X = 1, \quad U_X = R.$$

*It can be supported on one invertible affine map.*

*Proof.* Take  $R$  distinct provenance records, each of weight one, and attach all of them to the same map  $\phi(x) = x + 1$ . Their map-class mass is  $R$ , while every atom has weight one. □

**Proposition 6.2** (A quotient does not create dispersion). *Canonical aggregation preserves the census and its centered energy exactly. In particular, no bound for  $\mathfrak{Y}_H^\circ$  or  $U_X$  follows from the quotient identity alone.*

*Proof.* Equation (5) is an equality with the original census. Therefore its  $\ell^2$  norm and centered energy are unchanged. [Proposition 6.1](#) shows that one quotient class may have arbitrary positive mass in the abstract information model. □

*Remark 6.3* (Physical status). The countermodels prove nonimplications between abstract ledgers. They do not establish polynomial concentration in the actual TPC coefficient and hence do not stop the physical positive route. A future theorem may use the exact classification in [lemma 2.2](#) to bound actual class multiplicity or weight.

## 7 Finite exact regression

The deterministic script `experiments/tpc105_map_quotient_audit.py` constructs finite weighted atom families over several prime fields and checks:

1. direct and quotient multiplier censuses agree;
2. the map classification in [lemma 2.2](#);
3. the direct cross ledger equals both  $\mathfrak{K}_H^\circ + \mathfrak{Y}_H^\circ$  and the profile formulas;
4. the concentration inequality (12); and
5. the mass-only and atom-cap counterexamples.

All arithmetic is exact. The certificate tests the finite reindexing and claim schema; it is not evidence for a growing distinct-map incidence estimate.

## 8 Route consequence and claim boundary

### Established

- Identical affine maps are classified exactly by the pair  $(A_p, m_p)$ , with physical integer recovery for  $m_p$ .
- The canonical quotient retains the input profile needed for the positive census and the full bag needed for source recovery.
- The TPC-100 cross ledger splits exactly into identical- and distinct-map terms.
- The identical-map term is controlled by the new class-mass ledger  $U_X$ , and [corollary 5.4](#) gives a precise sufficient door for TPC-106.

These are L0 algebraic identities and an L1 provenance-preserving reduction on the actual fixed- $h_0$  carrier.

### Not established

- No bound  $U_X \leq X^{o(1)}$  is proved for the actual carrier.
- The principal bounds isolated in TPC-104 remain unproved [\[5\]](#).
- The distinct-map target [\(14\)](#) is not proved.
- No signed affine Möbius cancellation, full physical reassembly, strict 1/400 endpoint closure, or L2 fixed-power saving is obtained.
- No parity breakthrough, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime conclusion is proved or implied.

The honest next step is width-weighted incidence on the genuine distinct-map quotient, with  $U_X$  carried as a separate concentration gate. If either has a literal polynomial obstruction, the response is to retain the signed filter before majorization, not to claim that the original signed coefficient is large.

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